

Optimising the treatment of HDV infection: Efficacy of bulevirtide, HBV-HDV interactions and the importance of statistical methods

To the Editor:

I am writing to express my appreciation for the recent publication of the study entitled "*Blocking viral entry with bulevirtide reduces the number of HDV-infected hepatocytes in human liver biopsies*"¹ in the *Journal*. This study significantly advances our understanding of the efficacy of bulevirtide against HDV infection and its role in HBV-HDV interactions. It exemplifies the importance of integrating rigorous statistical methods in medical research, providing valuable insights and establishing a solid foundation for future research directions. The demonstration of bulevirtide's effectiveness is particularly noteworthy, offering a promising avenue for personalized treatment strategies. The study's innovative approach and potential clinical applications underscore its significance in the field of Primary Sclerosing Fibrosis (PSF) treatment research. Our suggestions aim to further enhance the study's impact, rather than critique its fundamental contributions.

Firstly, regarding the experimental design, although the patient grouping and variable control were executed with rigorous clinical trial design, more detailed subgroup analyses would be beneficial. For instance, there was a notable difference in the reduction of HDV RNA levels between patients treated with 2 mg and 10 mg of bulevirtide, but this difference was not consistent across all patients. We suggest categorizing patients into subgroups based on their baseline HDV RNA levels (high, intermediate, and low) to investigate whether those with higher viral loads respond better to high-dose bulevirtide therapy. Additionally, categorizing patients based on liver function indices (e.g., ALT, AST, bilirubin levels) into normal and impaired liver function groups could provide insights into how liver function status affects response to bulevirtide therapy.²

Secondly, although the study indicates that bulevirtide has significant efficacy against HDV with limited effect on HBV, it is crucial to consider the complex interaction between HBV and HDV. The persistence of HBV may influence the clearance of HDV.³ Therefore, future studies should delve deeper into the HBV-HDV interactions, particularly the changes in viral kinetics during treatment.⁴

As a gastroenterologist and R language practitioner, I would like to offer some cross-disciplinary suggestions to further enhance the impact of this study:

1. Import the study data into the R environment for cleaning and preprocessing.
2. Analyze the relationships between variables using correlation matrices, focusing on the relationship between treatment variables and patient response.⁵
3. Use multiple regression analyses to assess treatment effects and identify variables significantly impacting HDV RNA reduction.⁶
4. Facilitate the presentation of clinical results to the readers.
5. Validate the stability of the model using cross-validation techniques and employ criteria like AIC/BIC for model selection, ensuring both predictive power and parsimony.⁷

In conclusion, this study exemplifies the integration of rigorous experimental design and comprehensive statistical analysis, highlighting the critical role of statistics in medical research. By addressing the aforementioned suggestions, the study's impact can be further amplified, contributing even more significantly to the field.⁸

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Conflict of interest

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Supplementary data

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References

Author names in bold designate shared co-first authorship



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- [1] **Allweiss L, Volmari A**, Suri V, et al. Blocking viral entry with bulevirtide reduces the number of HDV-infected hepatocytes in human liver biopsies. *J Hepatol* 2024;80:882–891.
- [2] **Wedemeyer H, Schlaphoff V**, Cornberg M, et al. *Hepatology* 2019;70(4):1343–1356.
- [3] **Heidrich B, Deterding K**, Volz T, et al. *Gut* 2021;70(6):1175–1182.
- [4] **Lempp FA, Urban S**, Einav S, et al. *J Hepatol* 2021;75(1):62–75.
- [5] **Venkatraman A, Gupta N**, Brown L, et al. *Stat Methods Med Res* 2020;29(5):1341–1356.
- [6] **Sun H, Zhou D**, Xu H, et al. *J Stat Comput Simul* 2021;91(3):489–509.
- [7] **Nguyen T, Liang L**, Zeng Q, et al. *Comput Stat Data Anal* 2018;124:68–80.
- [8] **Ong G, Parameswaran K**, Fernandez J, et al. *Int J Med Inform* 2019;129:57–66.